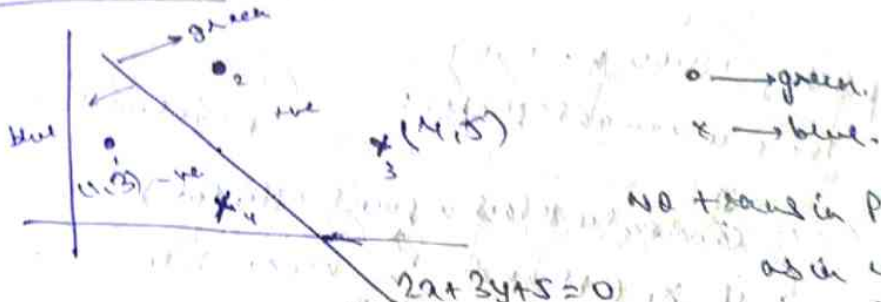


Logistic Regression



No + trans in P_1, P_2
as in correct
regions.

Transformation for P_3, P_1 .

Assume $P_3 = (4, 5, 1)$ P_3 in the region.

$$\begin{array}{ccc} 2 & 3 & 5 \\ - & 4 & 5 & 1 \\ \hline -2x + -2y + 4 = 0 \end{array} \rightarrow \text{New line that labels } (4, 5) \text{ in -ve region.}$$

$P_1 (1, 3)$ (P_1 in -ve region)

$$+ \begin{array}{ccc} 1 & 3 & 1 \\ \hline 3x + 6y + 6 = 0 \end{array} \rightarrow (1, 3) \text{ in -ve region.}$$

the $\rightarrow$ -ve (-kardu)
-ve $\rightarrow$ +ve (add)

in ML, (loss' = loss - η word)
 $\rightarrow 0.01$ (LR)

Algorithm:

x_0	$\log p(x_1)$	$\log(x_2)$	Placed (Y)
1	7.5	81	1
1	8.9	109	1
1	7.0	81	0

$$Ax + By + C = 0$$

$$W_0 + W_1 x_1 + W_2 x_2 = 0$$

$$W_0 x_0 + W_1 x_1 + W_2 x_2 = 0$$

$$\sum_{i=0}^2 W_i x_i = 0$$

$$W_0(1) + W_1(7.5) + W_2(81) \geq 0 \rightarrow 1$$

$$< 0 \rightarrow 0$$

Algo:

epochs $\rightarrow 1000, \eta = 0.01$

for i in range(epochs):

randomly select a point (student)

if $x_i \in N$ and $\sum_{i=0}^n w_i x_i > 0 \rightarrow \text{true}$

the region
but the point

the $\rightarrow$ ve IC-kards

$w_{\text{new}} = w_{\text{old}} - \eta [x_0 \ x_1 \ x_2]$
[$w_0 \ w_1 \ w_2$]

if $x_i \notin P$ and $\sum w_i x_i < 0$

$w_{\text{new}} = w_{\text{old}} + \eta x_i$

Simplified:

for i in epoch:

$(w_n = w_0 + \eta (y_i - \hat{y}_i) x_i)$

(real)

(predicted)

	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
$w_n = w_0 \leftarrow 1$	1	1	0
$w_n = w_0 \leftarrow 0$	0	0	0
$w_n = w_0 + \eta x_i \leftarrow 1$	1	0	1
$w_n = w_0 - \eta x_i \leftarrow 0$	0	1	-1

(Placed here, model says no)

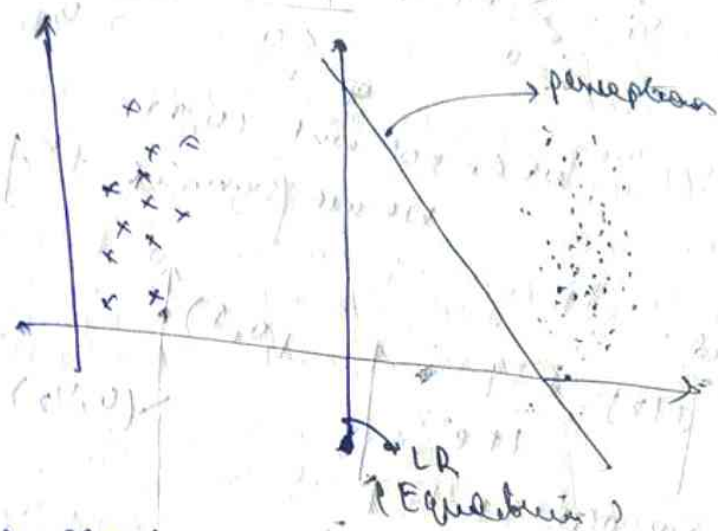
(No placed, model says yes)

the $\rightarrow$ -ve, (Subtract)

the point, -ve region

-ve $\rightarrow$ +ve (Add)

Perceptron vs Logistic Regression



As soon as all points are placed correctly, perceptron stops.

$$w_n = w_0 + \eta (y_i - \hat{y}_i) x_i$$

for misclassified (pull line)

In logistic regression, (line pull)

$$w_n = w_0 + \eta (y_i - \hat{y}_i) x_i \rightarrow \text{misclassified}$$

pull push $\rightarrow$ classified (line attains eq)

mag of push, pull $\rightarrow$ depends on distance of point and line.

now:

$$y_i \quad x_i$$

$$\begin{matrix} 1 & 1 \\ 0 & 0 \end{matrix} \rightarrow y_i - \hat{y}_i = 0 \Rightarrow w_n = w_0$$

$$\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \rightarrow w_n = w_0 + \eta x_i \quad (\text{As it is})$$

$$\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \rightarrow w_n = w_0 - \eta x_i$$

for LR, if we want push by classified points,

$$w_n \neq w_0 \quad \text{or} \quad (y_i - \hat{y}_i) \neq 0$$

const $\rightarrow$ model prediction

$$\hat{y}_i = \sum_{j=0}^2 w_j x_j \begin{matrix} \geq 0 \longrightarrow 1 \\ < 0 \longrightarrow 0 \end{matrix}$$

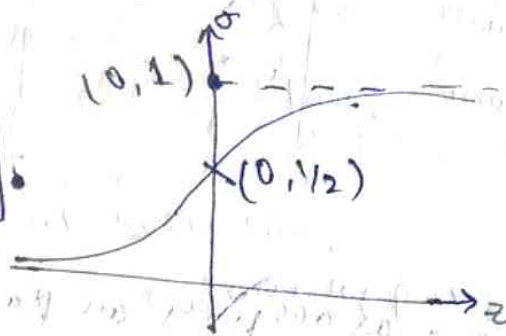
we do not want (0,1)

we use Sigmoid fn

Sigmoid:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

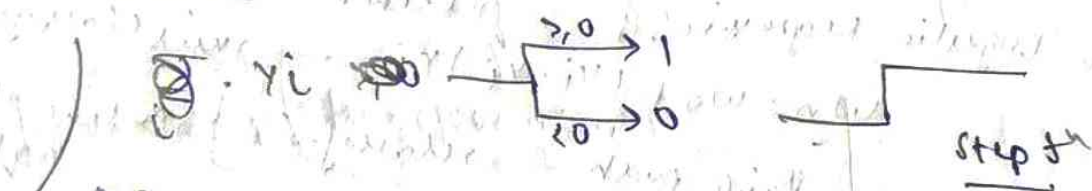
$$\begin{cases} z \rightarrow \infty, \sigma(z) \rightarrow 1 \\ z \rightarrow 0, \sigma(z) \rightarrow 0.5 \\ z \rightarrow -\infty, \sigma(z) \rightarrow 0 \end{cases}$$



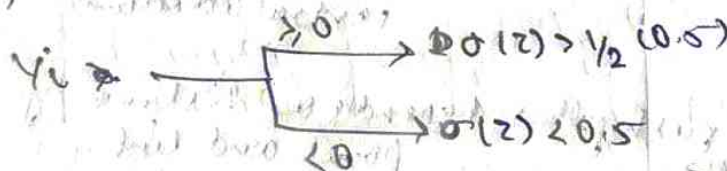
$$-\infty < z < \infty$$

$$0 < y < 1$$

earlier.



new,



Sigmoid fn

$\{7.5, 81\}$

find

$$\hat{y}_b = w_0 + w_1(7.5) + w_2(81) \longrightarrow z$$



$$w_0 = w_0 + \eta(0.2) \cdot 1$$

(push up) (Push line)

$$w_0 = w_0 + \eta(-0.65) \cdot 1$$

(Push line) (moving down)

$$w_0 = w_0 + \eta(0.7) \cdot 1$$

(Push) (moving up)

$$w_0 = w_0 + \eta(0.15) \cdot 1$$

(Push) (moving up)

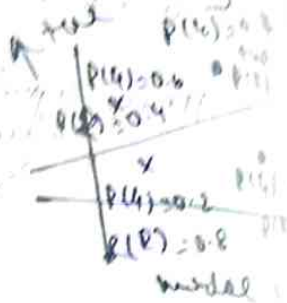
(Close the point to the line,

move the point,

loss function: (max likelihood)

• → green point
x → red point

$$\hat{y} = \sigma(z) \quad z = \sum_{i=0}^n w_i x_i$$

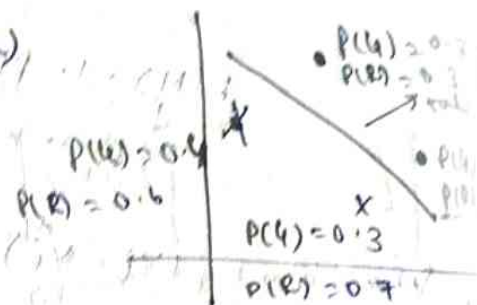


for model 1, (Prob of the green)

$$u_1 = (0.7 \times 0.4 \times 0.8 \times 0.4) = 0.089$$

mg:

$$u_2 = (0.6 \times 0.7 \times 0.6 \times 0.7) = 0.176$$



$u_2 > u_1$ (model 2 is better than model 1)
non likely hood

for max value of u_1 , find a line.

* For real world datasets, (OK + rows, not too many) may lead to a very small value.

hence we use $\log(u)$ & η

$$\log(\text{max}) = \log(0.7) + \log(0.4) + \log(0.4) + \log(0.8)$$

As all are -ve, to get rid of -ve sign,

$$-\sum \log(\text{max}) = -\log(0.7) - \log(0.4) - \log(0.4) - \log(0.8)$$

↓

cross entropy

$$\text{now } |\log(0.1)| > |\log(0.9)|$$

So cross entropy is never maximized,

cross entropy is minimized

$$L = \sum_{i=1}^n -y_i \log(\hat{y}_i) - (1-y_i) \log(1-\hat{y}_i)$$

Log loss loss or binary cross entropy soln

M1

$$P1, P(y) = 0.7$$

$$y_1 = 1$$

$$L_1 = -y_1 \log(\hat{y}_1) = -\log(0.7)$$

$$P2, P(y) = 0.6$$

$$y_2 = 0$$

$$L_2 = -1 \log(1-\hat{y}_1) = -\log(1-0.6) = -\log(0.4)$$

$$L_3 = -1 \log(1-0.2) = -\log(0.8)$$

$$L_4 = -1 \log(0.4) = -\log(0.4)$$

$$L_n = \sum L_i = \text{cross entropy}$$

Derivative of sigmoid

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\frac{d\sigma}{dx} = \frac{d}{dx} \left(\frac{1}{1+e^{-x}} \right) = \frac{d}{dx} (1+e^{-x})^{-1}$$

$$= - \frac{1}{(1+e^{-x})^2} \cdot e^{-x}$$

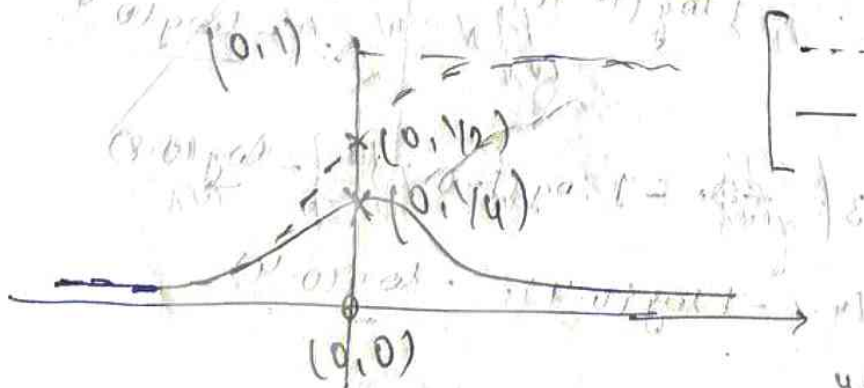
$$\frac{d\sigma}{dx} = \frac{1}{(1+e^{-x})^2} \cdot e^{-x}$$

$$\frac{e^{-x}}{(1+e^{-x})^2}$$

$$\sigma(x) \cdot \left(\frac{e^{-x}}{1+e^{-x}} \right)$$

$$\sigma(x) \left[1 - \frac{1}{1+e^{-x}} \right] = \sigma(x) [1 - \sigma(x)]$$

$$\boxed{\frac{d\sigma(x)}{dx} = \sigma(x) [1 - \sigma(x)]}$$



$\text{---} \rightarrow \sigma(x)$
 $\text{---} \rightarrow \sigma'(x)$

y intercept ($x=0$)

$$\sigma(0) = \frac{1}{2}$$

$$\frac{1}{2} (1 - \frac{1}{2}) = \frac{1}{4}$$

Gradient descent:

Row m , column n

$$\begin{array}{cccccc} x_{11} & x_{12} & \dots & x_{1n} & y_1 & \hat{y}_1 \\ x_{21} & x_{22} & \dots & x_{2n} & y_2 & \hat{y}_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} & y_m & \hat{y}_m \end{array}$$

$w_0, w_1, \dots, w_n \rightarrow$ weights
 $w_0 \rightarrow$ intercept

$$\sigma(w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n}) = \hat{y}_1$$

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix} = \begin{bmatrix} \sigma(w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n}) \\ \sigma(w_0 + w_1 x_{21} + w_2 x_{22} + \dots + w_n x_{2n}) \\ \vdots \\ \sigma(w_0 + w_1 x_{m1} + w_2 x_{m2} + \dots + w_n x_{mn}) \end{bmatrix}$$

$$\hat{y} = \sigma \left(\begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ 1 & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \right)$$

$$\hat{y} = \sigma(XW)$$

$$J = -\frac{1}{m} \sum_{i=1}^m \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]$$

$$\sum_{i=1}^m y_i \log(\hat{y}_i) = y_1 \log \hat{y}_1 + y_2 \log \hat{y}_2 + \dots + y_m \log \hat{y}_m$$

$$\begin{bmatrix} \log \hat{y}_1 \\ \vdots \\ \log \hat{y}_m \end{bmatrix}$$

$$[y_1 \quad y_2 \quad \dots \quad y_m] \log \left(\begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_m \end{bmatrix} \right)$$

$$= Y \log(\hat{Y})$$

$$L_1 = Y \log(\sigma(xw))$$

$$L_2 = (1-Y) \log(\sigma(xw))$$

$$L = \frac{1}{n} [Y \log \hat{Y} + (1-Y) \log (1-\hat{Y})]$$

$$\text{where } \hat{y} = \sigma(x \cdot w)$$

find w by GD.
minimise L :

for i in epochs:

$$w = w - \eta \frac{\Delta L}{\Delta w} \longrightarrow \text{GD.}$$

$$\frac{\Delta L}{\Delta w} = \left[\frac{\partial L}{\partial w_0}, \frac{\partial L}{\partial w_1}, \dots, \frac{\partial L}{\partial w_n} \right]$$

$$\frac{dL}{dw} (y \log \hat{y}) = y \frac{d}{dw} \log(\sigma(xw))$$

$$= \frac{y}{\hat{y}} \frac{d}{dw} (\sigma(xw))$$

$$= \frac{y}{\hat{y}} \sigma(xw) [1 - \sigma(xw)] \frac{d}{dw} (xw)$$

$$= \frac{y}{\hat{y}} \hat{y}(1-\hat{y}) \cdot x = \boxed{y(1-\hat{y})x}$$

$$\frac{d}{dw} [(1-y) \log(1-\hat{y})] = \cancel{(1-y)} \frac{d}{dw} \cancel{\sigma(1-xw)}$$

$$= \cancel{\frac{1-y}{1-\hat{y}}} \cdot \cancel{\sigma(1-xw) [1-\sigma(1-xw)]} \cdot \cancel{(-x)}$$

$$= \cancel{\frac{y-y}{1-\hat{y}}} \frac{(1-y)}{1-\hat{y}} \frac{d}{dw} \cancel{\sigma(1-xw)} [1-\hat{y}]$$

$$= \frac{-(1-y)}{(1-\hat{y})} \frac{d}{dw} \sigma(xw)$$

$$= - \frac{(1-y)}{1-\hat{y}} \hat{y}(1-\hat{y}) (x)$$

$$= \boxed{-\hat{y}(1-\hat{y})x}$$

$$\frac{dL}{dw} = -\frac{1}{w} [y(1-\hat{y})x - \hat{y}(1-y)x]$$

$$= -\frac{1}{w} [y(1-\hat{y}) - \hat{y}(1-y)]x$$

$$= -\frac{1}{w} [y - y/\hat{y} - \hat{y} + y/\hat{y}]x$$